Omkar Vikas Bhede | **193100065**
Mechanical Engineering | **M.Tech.**
Indian Institute of Technology, Bombay | **Gender: Male**
Specialization: Manufacturing Engineering | **DOB: 28-10-1996**

Examination	University	Institute	Year	CPI / %
Post Graduation	IIT Bombay	IIT Bombay	2021	**8.5**
Graduation	Pune University	VIT Pune	2018	**8.42**
Graduation Specialization: Mechanical Engineering				

Scholastic Achievements

- Awarded **AA** grade in **6** courses during M.Tech in IIT Bombay. [*'21*]
- Secured **99.28** percentile in GATE among 7,11,000 candidates. [*'19*]
- Se cured first division with **HONS.** in B.Tech. (Mechanical) from VIT, Pune. [*'18*]
- Awardee of Central Sector Scheme of **Scholarship** by Department of Higher Education. [*'14*]

Start-ups and Projects

Founder & CTO | *codefurry.com* *[May'21- Sept'21]*

- Built **EdTech SaaS platform** to help developers streamline the process of learning new skills, building projects, interviews & resume building to secure challenging positions in reputable organizations.
- Built mobile friendly front-end with **React** and **tailwindCSS**.
- Built scalable **NodeJs** backend with **express**-framework.
- Used **MongoDB Atlas** NoSQL Database and **JWT** tokens to make application stateless.
- Deployed frontend on **AWS Amplify** and backend using **ec2 instances** with **load balancer** and **auto scaling group** to cater the variable demand.
- Used **AWS S3** bucket to store and retrive 500GB+ videos at lowest latency using **aws-sdk** in NodeJS
- Deployed Video streaming api with nodeJS on **AWS elastic beanstalk** and GitHub **CICD pipeline**.
- **reactJS** – react-router, tailwind, axios, lottie, joi, Hooks, redux
- **nodeJS** – aws-sdk, bcryotjs, express, jsonwebtoken, mongoose, nodemailer, razorpay, Winston, morgan
- **AWS** – Amplify, Elastic Beanstalk, EC2, ASG, ALB, Route 53, Lambda, CodePipeline
- **Other** – pm2, nginx, redis, jest.

Co-Founder | *hypelala.com* *[Mar'21- Apr'21]*

- Built ecosystem for influencer marketing with **WordPress**.
- Collaborated with **100+ influencers** from Instagram.
- Used **spark AR** to create customised Instagram filter to deliver the Marketing Campaigns

Project | *naan* [*Jan'21- Feb'21*]

- Built mobile app to cater **billing process** of small restaurants and cafés
- Used **Flutter** (Dart) to develop cross platform mobile application with google **firebase**.
- Integrated advanced features like auto menu update from photos, sending bill through SMS via twilio APIs.

Positions of Responsibility

Sports Secretary | *Hostel 14, IIT Bombay* [*Jan'20-till date*]

- Inspired **100+** Hostel members to participate in Inter Hostel General Championship.
- Worked with team of 5 members and 200+ hostel members to get **6th rank** in Inter Hostel General Championship.
- Worked with the hostel council to **organize cultural events** in the hostel, like Dahi-handi, Holi, Lohri, Gala dinners.

Interview Coordinator | *The Placement Cell, IIT Bombay* [*Dec'19, Dec'20*]

- Coordinated with a team of **250+** members for interviews of **1600+** students.
- Assisted in conducting Pre-placement Talks and Tests for **15+** firms.

Tech-Connect Coordinator | *Techfest, IITB Bombay* [*Dec'19*]

- Coordinated with a team of **110+** members for the planning and execution of the event.
- Collaborated with **20+** exhibitors and facilitated their requirements.

Teaching Assistant | *Engineering Drawing and Graphics Lab* [*Jan'20-May'20*]

- Mentored **120+** undergraduate students to help build their skills in Engineering Drawing by guiding them in using software like **AutoCAD** and **SolidWorks**.
- Assisted in conducting lab sessions and provided support to conduct semester exams and helped students by solving their doubts and clearing key concepts related to the subject.

Extra-Curricular Activities and Achievements

Sports	• Participated in a **Football** Championship in **Hostel GC**. • Awarded **Bronze medal** in District Level **Judo** Tournament. • Participated in **Cyclothon 2019**, conducted by **Techfest**, IIT Bombay • Participated in **Half Marathon**.	[*'20*] [*'12*] [*'19*] [*'19*]
Organizational	• Co-ordinated College level **design competition organised** under SAE collegiate club,VIT. • H14 Hostel **Council member**.	[*'17*] [*'20*]
Technical	• Coursera – Achieved **97.3%** in 'Introduction to Programming in IOT', by Ivan Harris, UCL. • Undergone training program In **CATIA V5** by CADD Centre Pune.	[*'19*] [*'17*]
Others	• Former member of **SAE**- Society of Automotive Engineers. • **Hobbies** – Cycling, Playing Football, Gardening.	[*'18*]

Certification

- The Ultimate HTML/CSS Mastery
- The Ultimate JavaScript Mastery
- Node JS: Advanced Concepts
- Mastering React
- The Complete Node.js Course
- Introduction to Machine Learning

Course Projects

Computer-Aided Design of Micro-milling tools using generative design | M.Tech Project [*Jul' 20-till date*]
Guide: Prof. SS Pande, Department of Mechanical Engineering, IIT Bombay

- Modelled flat end mill and analysed using ADMAS.
- Using genetic algorithms to performed **wear rate** calculations.
- To use generative design methods in generated **tool geometries** to reduce modelling time.
- Interlinking of tool geometry designing formulation with **FEM analysis** methods.

Automating Colling Chamber o 3D Ice Printer | *Advisor: Prof. KP karunakaran* [*Jan'20-Mar'20*]

- Designed economical CO_2 based 3D Ice printer using **feedback control** loop.
- Flow simulations of CO_2 using **ANSYS-Fluent** are visualised.
- Flow rated are validated using **MATLAB** code.

Path planning in 3D environment with dynamic Obstacles | *Advisor: Prof. SS Pandey* [*Oct'19-Dec'19*]

- Reviewed A* algorithm and proposed a **new algorithm** based on object fields.
- Devised a MATLAB code to demonstrate the algorithm and **visualise the 3D** view of the path traversed.
- Outcomes: Robust Algorithm, Shortest path for dynamic obstacle, Quick computation ability, flexibility to user.

Application of Swarm Robotics in Military | B.Tech Project [*Jan'18-May'18*]
Guide: Prof. Santosh Joshi, Department of Mechanical Engineering, IIT Bombay

- Optimized the path of military tanks by using the collective behaviour of **decentralized** robots.
- Used **FireBird V** with **ZigBee** module to decrease the response time.
- Designed and Modelled mechanical gear **gripper** using 3D printer to increase quality and productivity of mechanism.

Formula Student India | Competition run by the Society of Automotive Engineers [*Mar'15-Jan'16*]

- Designed and Automated an **intake** system for uniform distribution of air-fuel charge.
- Designed and Automated an **exhaust** system with baffles for restricting noise below 100db.
- Worked with a team of **3** members on the engine and electronics of the vehicle.

Arduino based home automation | *B.Tech Course Project* [*Jan'16-May'16*]

- Controlling home appliances from **anywhere** in the world using a cellular connection.
- Decreased **cost to half** compared with Wi-Fi connected home automation.

Internships

- Completed 4 months internship in "**The Entrepreneurship Network**" as a **React Development -Associate**.
  Built various react projects like emoji search, calculator & website landing page for TEN network.
- Completed 2 months internship in Gear factory **of Mahabal Industries Pvt. Ltd.**, Miraj, Maharashtra.
  Worked on Designing and Manufacturing of Clampers of **Autoloader facility** in Kapp Machine

Technical Skills

• **Front-end**: **ReactJS**, ES7, Webpack2, Lodash, styled-components, Redux, HTML5, CSS, Bootstrap, Tailwind, Flutter
• **Back-end**: **NodeJS**, Firebase, jest, **ExpressJS**, Nginx, Redis, pm2
• **AWS**: Amplify, Elastic Beanstalk, EC2, ASG, ALB, Route 53, Lambda, CodePipeline, Certificates